Ahmed Thabet

Engineering Manager - Distributed Cloud Infrastructure & Platform Engineering

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Summary

Engineering leader with 10+ years building decentralized cloud infrastructure, custom operating systems, and developer platforms: from bare-metal OS design to Kubernetes orchestration at scale. 6+ of those years leading engineering teams, 3+ as an engineering manager, currently running multiple teams on a peer-to-peer cloud spanning thousands of nodes across 60+ countries.

Experience

Engineering Manager, Codescalers, Jan 2023 - Present

- Manage multiple engineering teams building the ThreeFold Grid, a decentralized P2P cloud spanning thousands of nodes across 60+ countries.
- Own technical strategy, architecture, roadmap, and delivery across the full stack: a custom Linux-based OS, multi-language SDKs, a Terraform provider, and web dashboards.
- Run hiring, onboarding, and mentoring; set code review, testing, and CI/CD standards (GitHub Actions, Forgejo).

Software Development Team Lead, Codescalers, Jan 2020 - Jan 2023

- Led development of the Jumpscale cloud automation framework, the 3Bot virtual system administrator, and the Virtual Data Center (VDC) platform.
- Drove system architecture and hands-on development while owning code quality and on-time delivery for the team.

Software Engineer, Codescalers, Jan 2016 - Jan 2020

- Designed and built cloud tooling for VM lifecycle management, container orchestration, and infrastructure automation.

Freelance & Contract Work

- **Data Analyst**, Otlob.com, Jan 2018 - Mar 2018: built an audio-fingerprinting pipeline over large-scale video datasets to measure marketing campaign reach.
- **Web Developer**, Orascom Telecom, Jan 2010 - Mar 2010: built an internal PHP/CodeIgniter application for server-room environmental monitoring, replacing manual Excel workflows.

Selected Projects

Operating Systems & Core Infrastructure

- *Zero-OS*: lead development of the custom Linux-based OS powering the entire ThreeFold Grid (Go).
- *ContentHive*: content-addressed storage system inspired by IPFS (Go).
- *0-Disk*: Network Block Device server with pluggable storage backends (Tarantool, Lua).
- *OpenVCloud*: monitoring, alerting, and hardware health-checks for a libvirt-based cloud platform (Python).

Platforms & Developer Tools

- *KubeCloud*: highly available Kubernetes platform on the ThreeFold Grid (Go, Vue.js, PostgreSQL).
- *VDC*: Virtual Data Center with managed Kubernetes clusters, S3 storage, and Prometheus/Grafana monitoring (Python, Stellar, Ansible).
- *VDC Marketplace*: one-click application deployment marketplace built on top of VDC.
- *llmrouter*: OpenAI-compatible LLM gateway with prepaid billing, provider routing and failover, and per-key policy (Go, PostgreSQL, Redis).
- *Diwan*: self-hosted AI staff that delivers real Word, Excel, and PowerPoint files with native Arabic right-to-left, scheduled routines, approvals, and MCP connectors (Go).
- *Hero Agent*: autonomous agent orchestrating goals, workers, and tools; its lean coding mode uses 1.7-5.3x fewer input tokens and benchmarked fastest of eight agents on open models (Rust).
- *Hero Researcher*: RAG research and background-check engine that flags unsourced claims (Rust).
- *Hero SafePad*: zero-knowledge encrypted editor; the key lives only in the link, the server stores ciphertext (TypeScript).
- *Terraform Provider*: Infrastructure-as-Code plugin for provisioning on the ThreeFold Grid (Go).
- *TFRobot*: mass-deployment tool for large VM fleets (Go).

- *Gridify*: Heroku-like deploy tool for the ThreeFold Grid (Go).
- *Farmerbot*: power-management and operations automation for grid farmers (Go).

SDKs & Frameworks

- *Go SDK*: primary SDK enabling delivery of multiple grid-based projects (Go).
- *TypeScript SDK*: web developer SDK for ThreeFold Grid integration (TypeScript).
- *JS-NG*: cloud automation framework (Python, Gevent).
- *JS-SDK*: Threebot application server built on JS-NG (Python).
- *EWF*: embeddable workflow framework for resilient, stateful multi-step processes (Go).
- *CRM*: CRM system for the ThreeFold Token Foundation (Python, Flask, GraphQL, PostgreSQL).
- *publishingtools*: web publishing framework for wikis, blogs, and websites (Crystal, Kemal, Svelte).

Infrastructure Tooling

- *AYS*: agentless Infrastructure-as-Code orchestration with an event system, background tasks, and service templates.
- *Go farmer*: cross-platform GUI for farm management (Go, Fyne). Additional: OAuth provider for Gogs, DMIDecode parser, GitHub-Gogs sync, and documentation generators for Jumpscale.

Open Source

Networking & Infrastructure

- uvm: runs OCI images as KVM microVMs with a Docker-like CLI, compose, and a JSON-RPC daemon (Go).
- chvm: Docker-like CLI running OCI images as microVMs on Cloud Hypervisor, with virtiofs volumes, vsock exec, and TAP/Mycelium networking (Rust).
- *castor*: content-addressed, commit-based object store; atomic tree commits, FastCDC/BLAKE3 dedup, and capability tokens instead of presigned URLs (Go).
- ewf: embeddable workflow engine with persisted state, crash resumption, retries, and queue-based execution (Go).
- tcprouter: TCP router with SNI-based routing (Go).
- tfgateway: ThreeFold gateway built on tcprouter and the CoreDNS Redis plugin.
- coredns-redis: Redis plugin for CoreDNS; raqeeb: PID 1 replacement and alternative to Foreman.

Python Libraries

- pygundb: GUN.js database server and client.
- python-sonic-client: client for the Sonic search backend.
- pyparsec: parser combinator library; python-pcre: PCRE bindings via ctypes; flask_itsyouonline: OAuth2 middleware for Flask.

Nim Ecosystem

- nim-servy: async web framework; nim-redisclient: Redis client and parser; nim-assets: bundle assets into a single binary.

Tools & Applications

- SafeBrowser: remote browser isolation streaming headless Chrome over CDP, with audio, mobile touch, and SSRF protection (TypeScript, Go).
- vscode-capnp: Cap'n Proto extension for VS Code; helloED: pluggable, fast PyGTK text editor.
- stellargui: lightweight Stellar wallet; raml2spore: generate SPORE clients from RAML specs.
- *murag*: offline RAG over personal notes with cited answers (Python); Pida IDE: RegexToolkit and Pastebin plugins.

Publications & Translations

- *Nim Days*: book on practical applications and libraries in Nim.
- *Python: Extensive Guide*: Arabic programming guide.
- *Introduction to Ruby*: Arabic programming guide.
- *Translations*: PyQt4, PyGTK, Mono IronPython, and Cairo tutorials translated into Arabic.

Skills

- **Leadership**: team management, hiring, mentoring, technical strategy, roadmap and delivery ownership, architecture review.

- **Languages**: Go (Golang), Python, Rust, TypeScript, C, Nim, Crystal, Bash.
- **Infrastructure**: Linux, Kubernetes, K3s, Helm, Terraform/OpenTofu, Ansible, Docker, Prometheus, Grafana, CoreDNS, Nginx, Traefik.
- **Data**: PostgreSQL, Redis, MySQL, MongoDB, InfluxDB.
- **APIs & protocols**: REST, GraphQL, gRPC, Cap'n Proto.
- **Frontend**: Vue, Svelte, React.

Education

- **BSc in Electrical Engineering**, Helwan University, Cairo, Egypt, 2014.
- **Mandatory military service**, Egyptian Armed Forces, 2015 - 2016.
- **Languages**: Arabic (native), English (business proficiency).